Nuclear Physics: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

15 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Nuclear Physics: worked solutions

Solution 1 — Read and transform a nuclide. Return →

$Z = \mathbf{92}$ and $N = 238 - 92 = \mathbf{146}$. Alpha decay gives $^{234}_{90}\mathrm{Th}$. Beta-minus decay of the original gives $^{238}_{91}\mathrm{Pa}$ plus an electron and antineutrino. Gamma emission changes the nucleus's energy state, not A or Z.

Check. Charge and nucleon number balance in each transformation.

Solution 2 — Weigh the binding. Return →

Using two hydrogen atoms cancels the two electrons in neutral helium: $\Delta m = 2m_H + 2m_n - m_{He} = \mathbf{0.03037664u}$. Multiplying by 931.494 MeV/u gives $B = 28.296$ **MeV**, or **7.074 MeV per nucleon**.

Check. Binding energy is positive because the bound atom is lighter than its separated constituents.

Solution 3 — Four hydrogens to helium. Return →

The atomic-mass difference is $4m_H - m_{He} = 0.02869687u$, giving $Q = 26.731$ **MeV**. Efficiency is $\Delta m/(4m_H) = \mathbf{0.00712}$, about 0.71%. The full weak process emits two positrons (which annihilate with electrons) and two neutrinos; atomic masses account for the electron bookkeeping. Neutrinos carry away part of the energy, so the photon/thermal share is smaller.

Check. Do not add positron masses again when the atomic-mass convention already balances the electrons.

Solution 4 — A fission fuel bill. Return →

One kilogram contains $1000/235$ mol, or 2.563×10^{24} nuclei. Each fission gives $200\,\mathrm{MeV} = 3.204 \times 10^{-11}$ J. Thus energy per kilogram is 8.21×10^{13} **J/kg**. A 1 GW thermal reactor uses 8.64×10^{13} J/day, corresponding to **1.05 kg fissioned per day**.

Check. This is thermal fuel consumption, not electrical output; real fuel loading and burnup differ.

Solution 5 — Three half-lives. Return →

24 days is three half-lives. Surviving fraction $= 2^{-3} = \mathbf{1/8} = \mathbf{12.5\%}$. Activity follows the same factor: **10 MBq**.

Check. Activity is proportional to the number remaining.

Solution 6 — A radiocarbon age. Return →

$t = -T_{1/2}\ln(0.72)/\ln 2 = 5730(0.4741) = \mathbf{2716\ yr}$ under the stated closed-system assumption.

Check. Less than one half-life has elapsed because more than half remains. Real radiocarbon calibration accounts for atmospheric variation and contamination.

Solution 7 — From atoms to becquerels. Return →

$\lambda = \ln 2/(5.27 \times 3.15576 \times 10^7) = 4.168 \times 10^{-9}$ s^{-1}. Initial activity $A_0 = \lambda N =$ $\mathbf{4.17 \times 10^6\ Bq} = 4.17$ MBq. After two half-lives, activity is one quarter: **1.04 MBq**.

Check. Convert the half-life to seconds before reporting becquerels.

Solution 8 — A daughter grows and decays. Return →

Idea. Daughter number peaks when production equals daughter decay: $\lambda_p N_p = \lambda_d N_d$. With half-lives 10 and 20 h, $\lambda_p = 2\lambda_d$. The two-member solution peaks at $t = \ln(\lambda_p/\lambda_d)/(\lambda_p - \lambda_d) = \mathbf{20\ h}$. Then

$N_p = N_0/4$ and $N_d = \mathbf{N_0/2}$. The activities are equal at that instant.

Check. The daughter is more numerous but decays more slowly per nucleus. This is not secular equilibrium because the parent is shorter-lived.

Solution 9 — A reaction threshold. Return →

Initial atomic mass exceeds final by $-0.00127953u$, so $Q = -1.192$ **MeV**. The reaction is endothermic. The stationary-target threshold is approximately $1.192(1 + 4/14) = \mathbf{1.53\ MeV}$.

Check. The threshold exceeds $|Q|$ because momentum conservation leaves kinetic energy in the final centre of mass. Atomic electrons balance: 7+2 on the left and 8+1 on the right.

Solution 10 — Count solar neutrinos. Return →

$26.2\,\mathrm{MeV} = 4.198 \times 10^{-12}$ J. Helium production rate $L_\odot/E = \mathbf{9.12 \times 10^{37}\ s^{-1}}$. Two neutrinos per reaction give 1.82×10^{38} s^{-1}. At 1 AU, flux $= 2\dot{N}/(4\pi d^2) = \mathbf{6.49 \times 10^{14}\ m^{-2}\,s^{-1}}$, or 6.49×10^{10} cm^{-2} s^{-1}.

Check. This is an all-flavour production estimate; oscillations change the electron-neutrino fraction detected at Earth.

Solution 11 — Fusion energy sharing. Return →

The mass difference is 0.01888289u, giving $Q = 17.589$ **MeV**. With negligible initial momentum, the two products have equal and opposite momenta. Nonrelativistically their kinetic energies divide inversely with mass: neutron gets about $4/5$ and alpha $1/5$. Thus $K_n \simeq 14.1$ **MeV** and $K_\alpha \simeq 3.5$ **MeV**.

Check. The lighter neutron carries most of the kinetic energy. The energies sum to Q.

Solution 12 — Why cooling continues. Return →

Fission products are radioactive and continue decaying after the neutron chain reaction stops. Their decay energy appears as **decay heat**, initially significant and then decreasing. Cooling must continue even when the reactor is shut down.

Check. Stopping fission does not stop all nuclear energy release.

Solution 13 — Parent and daughter. Return →

A 1:3 ratio means one quarter of the original parent remains. That is two half-lives, so the age is **24 years**. Initial daughter would make the sample appear older if not corrected; loss or gain of either isotope would also break the closed-system clock.

Check. Parent plus daughter is conserved only under the stated assumptions.

Solution 14 — Secular equilibrium. Return →

In secular equilibrium, $A_d \simeq A_p$. Since $A = \lambda N$, $N_d/N_p = \lambda_p/\lambda_d = T_d/T_p = 10/10^9 = \mathbf{10^{-8}}$. The daughter has far fewer nuclei but each decays much faster, so the total rates match.

Check. Equilibrium takes several daughter half-lives to establish and assumes the parent changes negligibly on that timescale.

Solution 15 — The Sun's nuclear clock. Return →

Available energy is $0.1 \times 0.007 M_\odot c^2$. Dividing by $L_\odot$ gives 3.27×10^{17} s $= \mathbf{1.04 \times 10^{10}\ yr}$. The gravitational Kelvin–Helmholtz scale is only about 3×10^7 yr.

Check. Nuclear burning provides a reservoir hundreds of times larger. The estimate assumes constant present luminosity and a fixed burnable fraction; a stellar evolution model refines it.